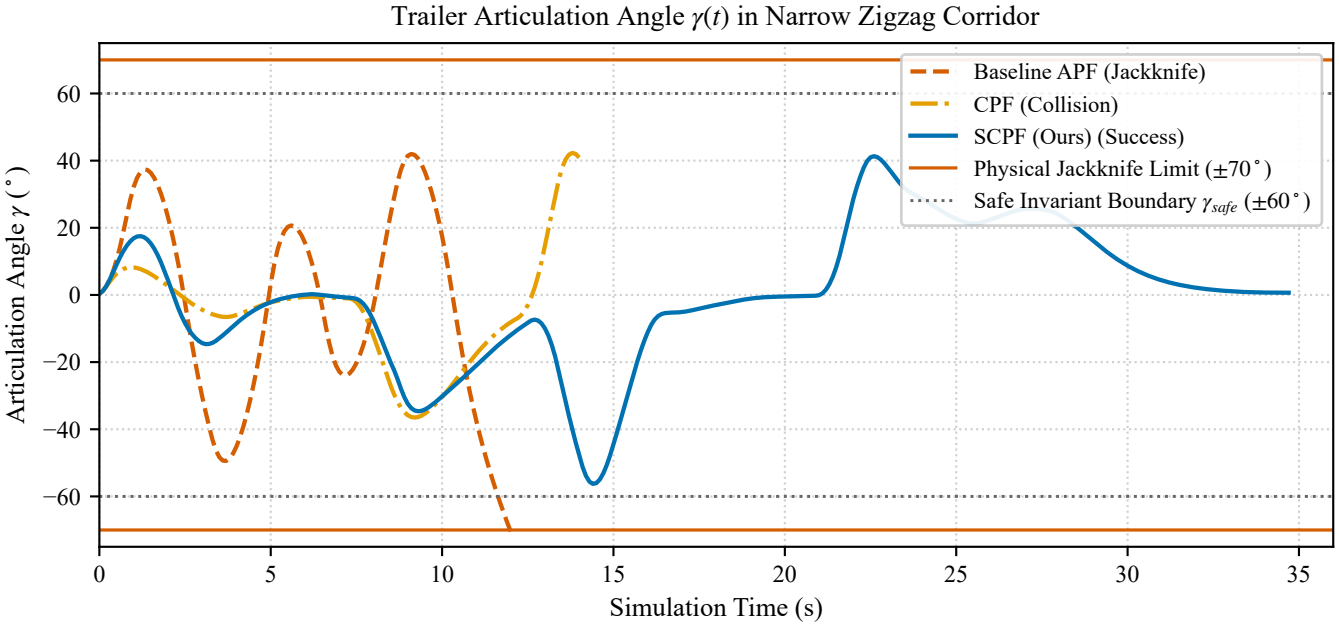

Trailer Articulation Angle $\gamma(t)$ in Narrow Zigzag Corridor
Baseline APF (Jackknife)
CPF (Collision)
SCPF (Ours) (Success)
Physical Jackknife Limit (±70°)
Safe Invariant Boundary $\gamma_{safe}$ (±60°)
Articulation Angle $\gamma$ (°)
Simulation Time (s)
60
40
20
0
−20
−40
−60
0
5
10
15
20
25
30
35